

Q#1: Determine the total running time and asymptotic upper bound.

b#1) Marks (4) <pre>int f2(int N){ for(int i=1; i<=N; i++) for(int j=1; j<=i; j++) sum++; return N; }</pre> Solution i=1 => j=1 i=2 => j=1,2 i=3 => j=1,2,3 ... i=N => j=1,2,3,...,N sum of arithmetic series: $1+2+3+\dots+N$ $N*(N+1)/2 = N^2$	b#2) Marks (4) <pre>void getData(int arr[], int N){ for(int i=1; i<N; i*=2){ for(int j=1; j<=i; j++) cin >> arr[i]; } }</pre> Solution i=1 => j=1 i=2 => j=1,2 i=4 => j=1,2,3,4 i=8 => j=1,2,3,4,...,8 ... i=N => j=1,2,3,4,...,N sum of geometric series: $1+2+4+8+\dots+N$ $1*(2^{\log_2 N}-1) / (2-1)$ Apply log rule to solve $2^{\log_2 N}$: $N^{\log_2 2} = N^1 = O(N)$
b#3) Marks (4) //you can assume that if statement will always execute. <pre>void f1(int N){ for(int i=1; i <= f2(N); i++){ if(arr[i] <= arr[i+1]) { int k = arr[i]; arr[i] = arr[i+1]; arr[i+1] = k; } } }</pre> Solution i=1 => N^2 i=2 => N^2 i=3 => N^2 ... i=N => N^2 $N^2 + N^2 + N^2 + \dots + N^2$ Since there are total N terms so $N*(N^2) = N^3$	b#4) Marks (8) including $O(f(N))$ <pre>int main(){ int N; cin >> N; for(int i=1; i<=N; i*=2){ getData(arr,N); f1(N); } return 0; }</pre> Solution: i=1 => $N + N^3$ i=2 => $N + N^3$ i=4 => $N + N^3$ i=8 => $N + N^3$... i=N => $N + N^3$ Since there are total $\log_2 N$ terms so $\log_2 N*(N + N^3) = N^3*\log_2 N$

a) for(i=1; i<=N; i++) { for(j=1; j<=i; j*=2) { stmt; } }	c) for(i=1; i<=N; i*=2) { for(j=1; j<=i; j*=2) { stmt; } }
d) //k = logN for(i=1; i<=N; i+=k) { stmt; }	e) for(i=1; i*i<=N; i++) { for(j=1; j<=N; j*=2) { stmt; } }
f) for(i=1; i<=N; i++) { for(j=1; j<=i*i; j++) { stmt; } }	g) k = sqrt(N) c = logN for(i=1; i*i<=N*N; i++) { for(j=1; j<=N; j+=k) { for(k=1; k<=N; k+=c) { stmt; } } }
h) for(i=1; i<=N; i++) { for(j=1; j<=i; j++) { for(k=1; k<=j; k++) { stmt; } } }	